CARTSHARE

BUILDING A COLLABORATIVE SHOPPING ECOSYSTEM

Project Type	Responsive front-end web application
Technology	HTML5 · CSS3 · JavaScript · Browser Storage
Application Views	Overview · Shared Cart · Activity Log · Group Receipt
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Submission	CartShare internship project

PROJECT REPORT

1. Project Problem Statement

In shared living or working environments such as student dormitories, office teams and travel groups, coordinating joint purchases can become confusing. People often use separate chat messages, notes or spreadsheets to record requirements. This fragmented process can cause missed items, duplicate purchases and repeated follow-up.

Key problems addressed:

- **Communication gap:** Participants do not have one common, continuously updated shopping list.
- **Delivery threshold:** Groups may need to coordinate purchases to reach a minimum order value for free delivery.
- **Low transparency:** It can be difficult to see who added or changed an item and to review the estimated group cost.

2. Project Objectives

The objective is to evolve a static front-end prototype into a responsive collaborative shopping experience. The application allows users to create or join a room using a code, maintain a common cart, see participant activity and generate an itemized printable receipt.

3. Proposed Solution and Main Features

Feature	Implementation
Room access	Room code creation and join form with input validation.
Shared cart	Add items with quantity and unit price; increase, decrease or remove items.
Activity log	Records cart actions with member label and timestamp.
Data persistence	Stores room, members, items and activity in browser localStorage.
Tab synchronization	Uses the browser storage event and BroadcastChannel to refresh open tabs.
Responsive interface	CSS Grid, Flexbox and media queries support mobile and desktop screens.
Receipt and printing	Creates an itemized group receipt and applies print-specific CSS.
Delivery target	Shows progress toward the ■75 target and an estimated delivery amount.

4. Tools and Technologies

HTML5: Semantic page structure, forms and navigation. **CSS3:** Responsive layout, visual styling, component states and print media rules. **JavaScript:** Form validation, cart operations, calculations, routing, activity logging and event handling. **localStorage:** Persistent browser-side room data. **BroadcastChannel / storage event:** Synchronization between open tabs using the same browser profile and origin.

5. Application Workflow

1. Open the Overview page and review the room code, item count and estimated total.
2. Create a new room or enter a room code to switch/join a room.
3. Add a product name, quantity and optional unit price to the shared cart.
4. Update quantities or remove items; the summary recalculates automatically.
5. Open the Activity Log to review actions and participant labels.
6. Open Group Receipt to review the itemized total and print or save it as PDF.

6. Testing and Success Criteria

The following checks should be performed before final portal submission:

Test	Expected result
Add item	New item appears and subtotal/total update.
Quantity controls	Quantity changes and line total recalculates.
Remove item	Item disappears and totals update.
Refresh page	Room and cart data remain available.
Two tabs	An update in one tab is reflected in the other tab.
Responsive layout	Navigation and cart remain usable on phone and desktop.
Print receipt	Receipt prints without navigation or dashboard elements.

7. Limitations and Future Scope

This is a front-end prototype. Browser localStorage and BroadcastChannel synchronize tabs that share the same browser profile and origin; they do not synchronize independent devices or browsers. A production version can add a backend database, real-time subscriptions, authentication, role-based access, cloud deployment and payment-splitting support. Delivery fees are illustrative estimates and not connected to a retailer.

8. Submission and Deployment Checklist

- Upload the complete project folder to a GitHub repository using the required naming convention: BatchID_FullName_CartShare.
- Deploy the static project using GitHub Pages, Netlify or Vercel.
- Test the deployed public URL on mobile and desktop, including two tabs on the same browser.
- Capture clear screenshots of Overview, Shared Cart, Activity Log and Group Receipt.
- Record a 2–5 minute walkthrough using VIDEO_DEMO_SCRIPT.md.
- Submit the live URL, repository URL, project files and this report together.

Important: Do not claim that a public deployment or cross-device backend exists until you have completed and tested those steps. The supplied project includes local run and deployment instructions.